

Oscar Valles

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SUMMARY

Computer Engineering graduate student with hands-on experience building AI-enabled applications using Python, LLMs, APIs, AWS, and cloud-native services. Background includes developing AI agents and automation workflows, designing REST-based systems, prototyping data-driven solutions, and deploying scalable applications across chatbot, fraud detection, and serverless machine learning projects. Experienced in translating business problems into working technical solutions, collaborating with stakeholders, and iterating quickly in project-based environments.

EDUCATION

The University of Texas at Dallas

Richardson, TX | Master of Science in Computer Engineering | Expected May 2026 | GPA: 3.69
University of Phoenix — Bachelor of Science in Information Technology

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, C++, Node.js
AI and Automation: LLMs, AWS Bedrock, AI agents, prompt design, NLP, automation workflows, structured multi-step systems
APIs and Frameworks: REST APIs, Flask, GraphQL, Boto3, Streamlit
Cloud and Platforms: AWS Lambda, API Gateway, DynamoDB, S3, Cognito, CloudFront, Bedrock
Development Practices: prototyping, testing, debugging, CI/CD, Git, Agile, Scrum

EXPERIENCE

Cloud Application Developer

Cloud Academy — Remote, TX | October 2024 to Present

- Built backend data pipelines with DynamoDB and GraphQL to support AI-enabled applications and real-time data.
- Supported development and testing of AI-integrated features, increasing user engagement by 25% through personalized recommendations and real-time updates.
- Collaborated with cross-functional teams to gather requirements, configure solutions, and document technical outcomes for deployment readiness.
- Implemented CI/CD workflows that reduced deployment time by 50% and supported 99.9% uptime across production environments.

PROJECTS

AWS AI-Powered Chatbot | Python, AWS Bedrock, Streamlit, Boto3

GitHub: github.com/ovalles2019/Bedrock-Chatbot

- Designed and developed a multilingual AI chatbot using AWS Bedrock and Anthropic Claude to support real-time question-answer interactions.
- Built API-connected workflows and AI agent logic to handle user queries, validate requests, and improve response reliability.
- Rapidly prototyped, tested, and iterated on chatbot features, improving user engagement by 40% through faster multilingual query resolution.

PotionWatch - Fraud Detection System | Python, Flask, Pandas, NumPy

HackUTD 2025 | GitHub: github.com/ovalles2019/fraud-detection-system

- Developed an automated fraud detection system for EOG Resources to identify discrepancies in transport documentation, suspicious tickets, and missing records across transactions.
- Built a RESTful API with statistical outlier detection and real-time ticket-matching logic to process operational data across 12 collection sites.
- Translated an audit and revenue-protection problem into a working solution that helped operations teams investigate anomalies faster.

Spam Detection Service / ML Cloud Deployment | Python, AWS, API Gateway, Lambda, S3, CloudWatch

GitHub: github.com/ovalles2019/ml-cloud-deployment

- Built and deployed a cloud-based inference workflow using AWS serverless services to support end-to-end spam classification requests.
- Tested and monitored latency, throughput, and error behavior with CloudWatch dashboards to improve performance visibility and release readiness.
- Documented deployment and troubleshooting procedures to support repeatable validation and stable operation.

LEADERSHIP

AWS Cloud Club, UTD — Core Team Member

- Organized and led technical workshops on cloud and AI topics, presenting technical concepts and project findings to 100+ attendees.
- Mentored students on AI and cloud career development, strengthening technical engagement across the club.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03) — Amazon Web Services
AWS Certified Data Engineer - Associate (DEA-C01) — Amazon Web Services